保护性暂停模式

定义（Guarded Suspension Design Pattern）

1. 某个结果需要在多线程之间传递，则可以让这些线程关联到一个对象 GuardedObject
2. 但是如果这个结果需要不断的从一个线程到另一个线程那么可以使用消息队列（见生产者/消费者）
3. 我们前面前面说的join、future采用的就是这个模式

如何实现

最简单的实现

1、首先编写一个简单的GuardedObject

```
package com.shadow.guarded;

import lombok.extern.slf4j.Slf4j;

@Slf4j(topic = "enjoy")
public class GuardedObject {
    private Object response;

    Object lock = new Object();

    /**
     * 加锁获取 response的值 如果response 没有值则等待
     * @return
     */
    public Object getResponse(){
        synchronized (lock) {
            log.debug("主线程 获取 response 如果为null则wait");
            while (response == null) {
                try {
                    lock.wait();
                } catch (InterruptedException e) {
                    e.printStackTrace();
                }
            }
        }
        return response;
    }


    /**
     * t1 给response设置值
     * @param response
     */
    public void setResponse(Object response) {
        synchronized (lock) {
```

```
            this.response = response;
            //设置完成之后唤醒主线程
            lock.notifyAll();
        }
    }
}
```

2、编写模拟耗时操作

```
package com.shadow.guarded;

import java.util.concurrent.TimeUnit;

public class Operate {

    public static String dbOprate(){
        try {
            TimeUnit.SECONDS.sleep(4);
        } catch (InterruptedException e) {
            e.printStackTrace();
        }
        return "result";
    }


}
```

3、编写测试类

```
package com.shadow.guarded;

import lombok.extern.slf4j.Slf4j;

@Slf4j(topic = "enjoy")
public class Test {
    public static void main(String[] args) {
        GuardedObject guardedObject = new GuardedObject();
        new Thread(() -> {
            String result = Operate.dbOprate();
            log.debug("t1 set完毕...");
            guardedObject.setResponse(result);
        },"t1").start();


        log.debug("主线程等待t1 set");

        Object response = guardedObject.getResponse();
        log.debug("response: [{}] lines",response);
    }
```

```
}
```

结果

```
17:36:15.856 [main] DEBUG enjoy - 主线程等待t1 set
17:36:15.859 [main] DEBUG enjoy - 主线程 获取 response 如果为null则wait
17:36:19.856 [t1] DEBUG enjoy - t1 set完毕...
17:36:19.856 [main] DEBUG enjoy - response: [result] lines

Process finished with exit code 0
```

超时实现

如果想要实现超时，那么在get的时候需要定义一个超时时间

```
public Object getResponse(long millis)
```

然后wait的不能无限的等待

```
lock.wait(millis);
```

继而结束while循环

```
public Object getResponse(long millis){
    synchronized (lock) {
        log.debug("主线程 获取 response 如果为null则wait");
        while (response == null) {
            try {
                lock.wait(millis);
                break;
            } catch (InterruptedException e) {
                e.printStackTrace();
            }
        }
    }
    return response;
}
```

分析1

这种做法的问题在于如果主线程被别人叫醒了；就会立马返回；比如超时时间是5s；但是在第2s的时候别人把主线程叫醒了，那么主线程会立马返回没有等足5s

所以需要设计一个经历时间；也就是从他wait到被别人叫醒中间一共经历了多少时间；判断这个时间是否符合超时;如果要计算这个经历时间必须知道开始时间和结束时间；

1、首先定一个开始时间等于当前时间 long begin = System.currentTimeMillis();

2、定一个经历时间 默认为0 long timePassed = 0;

3、判断是否满足条件，满足则返回结果不阻塞；不满足则然后进入while循环 首先计算等待时间（也就是wait的时间） millis-timePassed

4、判断等待时间是否小于0；小于0标识超时了直接结束while循环 返回不等待了

5、如果大于0 进入wait 这样就算提前被别人叫醒 也会在继续wait

最终代码实现

```
package com.shadow.guarded;

import lombok.extern.slf4j.Slf4j;

@Slf4j(topic = "enjoy")
public class GuardedObjectTimeOut {
    private Object response;

    Object lock = new Object();

    /**
     * 加锁获取 response的值 如果response 没有值则等待
     * @return
     */
    public Object getResponse(long millis){
        synchronized (lock) {
            //开始时间
            long begin = System.currentTimeMillis();
            //经历了多少时间 开始肯定是0
            long timePassed = 0;
            while (response == null) {
                long waitTime = millis-timePassed;
                log.debug("主线程  判断如果没有结果则wait{}毫秒",waitTime);
                if (waitTime <= 0) {
                    log.debug("超时了 直接结束while 不等了");
                    break;
                }
                try {
                    lock.wait(waitTime);
                } catch (InterruptedException e) {
                    e.printStackTrace();
                }
                //如果被别人提前唤醒 先不结束 先計算一下经历时间
                timePassed = System.currentTimeMillis() - begin;
                log.debug("经历了: {}", timePassed);
            }
        }
        return response;
    }


    /**
     * t1 给response设置值
     * @param response
     */
    public void setResponse(Object response) {
        synchronized (lock) {
            this.response = response;
            //设置完成之后唤醒主线程
            lock.notifyAll();
        }
    }
```

```
}
```

死锁

如果线程需要获取多把锁那么就很可能会发现死锁

```
package com.shadow.lock;

import jdk.nashorn.internal.ir.Block;
import lombok.extern.slf4j.Slf4j;

import java.util.concurrent.TimeUnit;

@Slf4j(topic = "enjoy")
public class LockTest {

    //定义两把锁
    static Object x = new Object();
    static Object y = new Object();

    public static void main(String[] args) {
        //线程1启动
        new Thread(()->{
            //获取x的锁
            synchronized (x){
                log.debug("locked x");
                try {
                    TimeUnit.SECONDS.sleep(1);
                } catch (InterruptedException e) {
                    e.printStackTrace();
                }
                synchronized (y){
                    log.debug("locked x");
                    log.debug("t1---------");
                }
            }

        },"t1").start();



        new Thread(()->{
            synchronized (y){
                log.debug("locked y");
                try {
                    TimeUnit.SECONDS.sleep(2);
                } catch (InterruptedException e) {
                    e.printStackTrace();
                }
                synchronized (x){
                    log.debug("locked x");
                    log.debug("t2---------");
                }
            }
```

```
        },"t2").start();
    }


}
```

活锁

不可避免 但是我可以错开他们的执行时间

```
package com.shadow.lock;

import lombok.extern.slf4j.Slf4j;

import java.util.concurrent.TimeUnit;

@Slf4j(topic = "enjoy")
public class LockTest1 {
    static volatile int count = 10;
    static final Object lock = new Object();


    public static void main(String[] args) {
        //t1线程对count一直做减法 直到减为0才结束
        new Thread(() -> {
            while (count > 0) {
                try {
                    TimeUnit.NANOSECONDS.sleep(2000);
                } catch (InterruptedException e) {
                    e.printStackTrace();
                }
                count--;
                log.debug("count: {}", count);
            }
        }, "t1").start();


        //t2线程对count一直做加法 直到加为20才结束
        new Thread(() -> {
            while (count < 20) {
                try {
                    TimeUnit.NANOSECONDS.sleep(2000);
                } catch (InterruptedException e) {
                    e.printStackTrace();
                }
                count++;
                log.debug("count: {}", count);
            }
        }, "t2").start();
    }
}
```

Lock--应用

特点:

1. 可打断，可重入
2. 可以设置超时时间
3. 可以设置为公平锁
4. 支持多个条件变量
5. 支持读写锁(单独的篇章来讲)

基本语法

```
// 获取锁
reentrantLock.lock();
try {
// 临界区
} finally {
// 释放锁
reentrantLock.unlock();
}
```

重入

```
package com.shadow.lock;

import lombok.extern.slf4j.Slf4j;

import java.util.concurrent.locks.ReentrantLock;

@Slf4j(topic = "enjoy")
public class LockTest3 {

    //首先定义一把锁
    static ReentrantLock lock = new ReentrantLock();

    public static void main(String[] args) {
        lock1();
    }


    public static void lock1() {
        lock.lock();
        try {
            log.debug("执行lock1");
            //重入
            lock2();
        } finally {
            lock.unlock();
        }
    }
```

```
    public static void lock2() {
        lock.lock();
        try {
            log.debug("执行lock2");
        } finally {
            lock.unlock();
        }
    }

}
```

可打断

```
package com.shadow.lock;

import lombok.extern.slf4j.Slf4j;

import java.util.concurrent.TimeUnit;
import java.util.concurrent.locks.ReentrantLock;

@Slf4j(topic = "enjoy")
public class LockTest4 {

    public static void main(String[] args) throws InterruptedException {
        ReentrantLock lock = new ReentrantLock();


        //t2首先获取锁 然后阻塞5s
        new Thread(()->{
            try {
                lock.lock();
                log.debug("获取锁----");
                TimeUnit.SECONDS.sleep(5);
                log.debug("t2 5s 之后继续执行");
            } catch (InterruptedException e) {
                e.printStackTrace();
            } finally {
                lock.unlock();
            }

        },"t2").start();


        TimeUnit.SECONDS.sleep(1);

        //t1加锁失败因为被t2持有
        Thread t1 = new Thread(() -> {
            try {
                lock.lockInterruptibly();
                log.debug("获取了锁--执行代码");
            } catch (InterruptedException e) {
                e.printStackTrace();
                log.debug("被打断了没有获取锁");
```

```
                return;
            } finally {
                lock.unlock();
            }
        }, "t1");
        t1.start();



        //由于t1 可以被打断 故而1s之后打断t1 不在等待t2释放锁了
        try {
            log.debug("主线程------1s后打断t1");
            TimeUnit.SECONDS.sleep(2);
            t1.interrupt();
        } catch (InterruptedException e) {
            e.printStackTrace();
        }

    }
}
```

```
t---线程
lock.lockInterruptibly();
标识可以打断

怎么打断

t.interrupt();
```

超时

```
package com.shadow.lock;

import lombok.extern.slf4j.Slf4j;

import java.util.concurrent.TimeUnit;
import java.util.concurrent.locks.ReentrantLock;

@Slf4j(topic = "enjoy")
public class LockTest5 {

    public static void main(String[] args) throws InterruptedException {
        ReentrantLock lock = new ReentrantLock();
        Thread t1 = new Thread(() -> {
            log.debug("t1启动---------");
            try {
                if (!lock.tryLock(2,TimeUnit.SECONDS)) {//嘗試获取锁,如果失败则返回
                    log.debug("拿不到鎖，返回");
                    return;
                }
```

```
            } catch (InterruptedException e) {
                e.printStackTrace();
            }
            try {
                log.debug("获得了锁");
            } finally {
                lock.unlock();
            }
        }, "t1");
        lock.lock();
        log.debug("主綫程获得了锁");
        t1.start();
        try {
            TimeUnit.SECONDS.sleep(3);
        } finally {
            lock.unlock();
        }

    }
}
```

多個條件

synchronized 中也有条件变量，就是以前讲的waitSet 不满足条件的线程进入waitSet；而Lock也有waitSet而且有多个

```
package com.shadow.lock;

import lombok.extern.slf4j.Slf4j;

import java.util.concurrent.locks.Condition;
import java.util.concurrent.locks.ReentrantLock;

@Slf4j(topic = "enjoy")
public class TestWait5 {

    static final ReentrantLock lock = new ReentrantLock();
    static boolean isPrettyGril = false; // 女人
    static boolean isMoney = false;//工资

    //没有女人的 waitSet
    static Condition waitpg = lock.newCondition();
    // 没有钱的waitSet
    static Condition waitm = lock.newCondition();


    public static void main(String[] args) throws InterruptedException {
        new Thread(() -> {
           try {
               lock.lock();
                   log.debug("有没有女人[{}]", isPrettyGril);
                   while (!isPrettyGril) {
                       log.debug("没有女人！等女人");
                       try {
                           waitpg.await();
```

```
                    } catch (InterruptedException e) {
                        e.printStackTrace();
                    }
                }
                log.debug("男女搭配干活不累；啪啪啪写完了代码");
            }finally {
                lock.unlock();
            }
        }, "jack").start();



        new Thread(() -> {
            try {
                lock.lock();

                log.debug("有没有工资[{}]", isMoney);
                while (!isMoney) {
                    log.debug("没有工资！等发工资");
                    try {
                        waitm.await();
                    } catch (InterruptedException e) {
                        e.printStackTrace();
                    }
                }
                log.debug("-----卧槽好多钱；啪啪啪写完了代码");

            }finally {
                lock.unlock();
            }
        }, "rose").start();



       Thread.sleep(1000);
        new Thread(() -> {
            try {
                lock.lock();
                isMoney = true;
                log.debug("钱来哦了");
                waitm.signal();
                isPrettyGril=true;
                waitpg.signal();

            }finally {
                lock.unlock();
            }
        }, "boss").start();
    }




}
```

读写锁

读读并发

读写互斥

写写互斥

读写锁读锁不支持条件

```
读锁的条件直接调用ReentrantReadWriteLock的 newCondition 会直接exception
public Condition newCondition() {
    throw new UnsupportedOperationException();
}
```

读写锁使用的例子

```
package com.shadow.lock;

import com.shadow.aqs.CustomSync;
import lombok.extern.slf4j.Slf4j;


import java.util.concurrent.TimeUnit;
import java.util.concurrent.locks.*;


@Slf4j(topic = "enjoy")
public class LockTest10 {

      static ReentrantReadWriteLock rwl = new ReentrantReadWriteLock();
      static Lock r = rwl.readLock();
      static Lock w = rwl.writeLock();


    public static void main(String[] args) throws InterruptedException {

        //读
        new Thread(()->{
            log.debug("read 获取 锁");
            r.lock();
            try {
                for (int i = 0; i < 10; i++) {
                    m1(i);
                }
            }finally {
                r.unlock();
            }


        },"t1").start();

        //写
        new Thread(()->{
            log.debug("write 获取 锁");
            w.lock();
            try {
```

```
                for (int i = 0; i < 20; i++) {
                    m1(i);
                }
            }finally {
                w.unlock();
            }

        },"t2");


        //读
        new Thread(()->{
            log.debug("write 获取 锁");
            r.lock();
            try {
                for (int i = 0; i < 20; i++) {
                    m1(i);
                }
            }finally {
                r.unlock();
            }

        },"t3").start();

    }


    public static void m1(int i){
            log.debug("exe"+i);
        try {
            TimeUnit.SECONDS.sleep(1);
        } catch (InterruptedException e) {
            e.printStackTrace();
        }

    }
}
```

读写支持重入但是只支持降级不止升级

```
package com.shadow.lock;

import lombok.extern.slf4j.Slf4j;

import java.util.concurrent.TimeUnit;
import java.util.concurrent.locks.Lock;
import java.util.concurrent.locks.ReentrantReadWriteLock;


@Slf4j(topic = "enjoy")
public class LockTest11 {

      static ReentrantReadWriteLock rwl = new ReentrantReadWriteLock();
```

```java
        static Lock r = rwl.readLock();
        static Lock w = rwl.writeLock();


    public static void main(String[] args) throws InterruptedException {

        new Thread(() -> {
            log.debug("read");
            w.lock();
            try {
                log.debug("read 已经获取");
                r.lock();
                log.debug("write 已经获取");
            } finally {
                r.unlock();
                w.unlock();

            }


        }, "t1").start();


    }
}
```

```java
//缓存
class CachedData {
    Object data;
    //判断缓存是否过期
    volatile boolean cacheValid;
    //定义一把读写锁
    final ReentrantReadWriteLock rwl = new ReentrantReadWriteLock();
    //处理缓存的方法
     void processCachedData() {
      rwl.readLock().lock();
      //如果缓存没有过期则调用 use(data);
      if (!cacheValid) {//要去load真实数据   set给缓存拿到写锁
        //释放读锁 因为不止升级 所以需要先释放
        rwl.readLock().unlock();


        rwl.writeLock().lock();
        try {
          //双重检查
          if (!cacheValid) {
            data = "数据库得到真实数据"
            cacheValid = true;
          }

          //更新缓存之后接着读取 所以先加锁
          rwl.readLock().lock();
        } finally {
          rwl.writeLock().unlock(); // Unlock write, still hold read
        }
      }
```

```
        try {
            //不管上面的if进不进都会执行这里
          //缓存可用
          use(data);
        } finally {
          rwl.readLock().unlock();
        }
      }
    }
  }
```

AQS框架

定义

1、全称是 AbstractQueuedSynchronizer

2、阻塞式锁和相关的同步器工具的框架；

3、AQS用一个变量（volatile state） 属性来表示锁的状态，子类去维护这个状态

3、getState、compareAndSetState cas改变这个变量

4、独占模式是只有一个线程能够访问资源

5、而共享模式可以允许多个线程访问资源（读写锁）

6、内部维护了一个FIFO等待队列，类似于 synchronized关键字当中的 Monitor 的 EntryList

7、条件变量来实现等待、唤醒机制，支持多个条件变量，类似于 Monitor 的 WaitSet

8、内部维护了一个Thread exclusiveOwnerThread 来记录当前持有锁的那个线程

功能

1、实现阻塞获取锁 acquire 拿不到锁就去阻塞 等待锁被释放再次获取锁

2、实现非阻塞尝试获取锁 tryAcquire 拿不到锁则直接放弃

3、实现获取锁超时机制

4、实现通过打断来取消

5、实现独占锁及共享锁

6、实现条件不满足的时候等待

自定义实现AQS框架

继承AQS 实现其主要方法

```
package com.shadow.aqs;

import java.util.concurrent.locks.AbstractQueuedLongSynchronizer;
```

```java
import java.util.concurrent.locks.Condition;

public class CustomSync extends AbstractQueuedLongSynchronizer {
    @Override
    public boolean tryAcquire(long acquires) {
        if (compareAndSetState(0, acquires)) {
            setExclusiveOwnerThread(Thread.currentThread());
            return true;
        }
        return false;
    }


    @Override
    protected boolean tryRelease(long arg) {

        if(getState() == 0) {
            throw new IllegalMonitorStateException();
        }
        setExclusiveOwnerThread(null);
        setState(0);
        return true;
    }

    @Override
    protected boolean isHeldExclusively() {
        return getState()==1;
    }


    public Condition newCondition() {
        return new ConditionObject();
    }
}
```

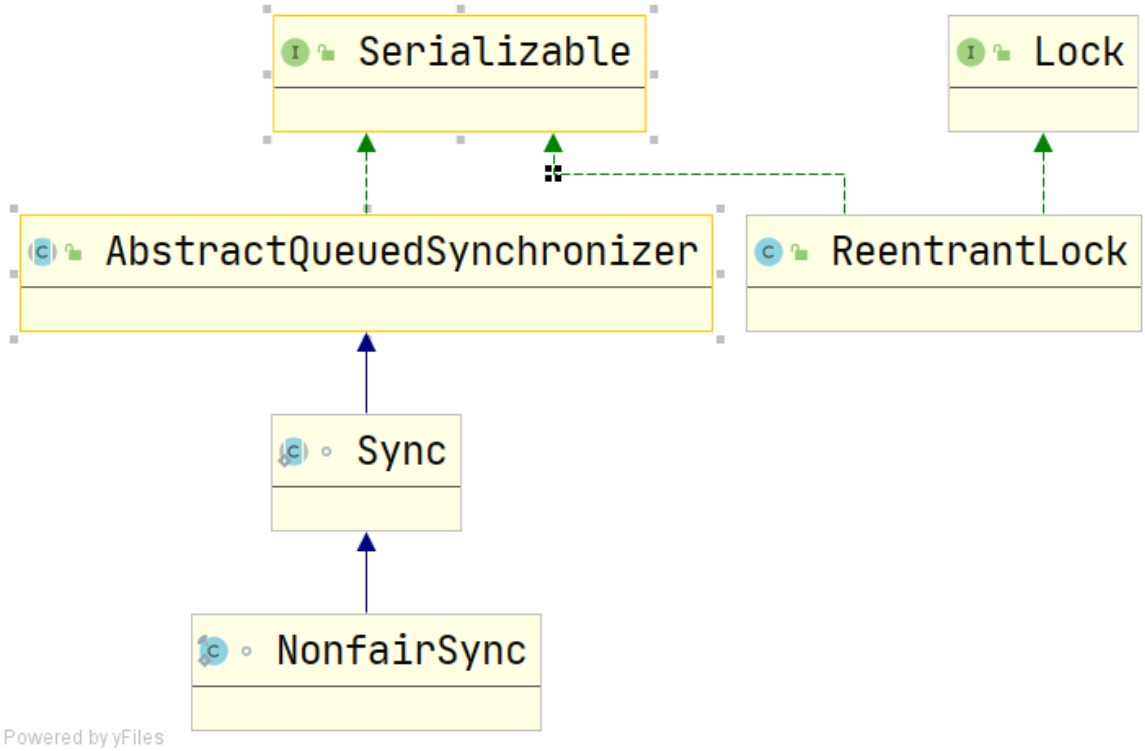
实现Lock接口实现加锁解锁

```java
package com.shadow.lock;

import com.shadow.aqs.CustomSync;
import lombok.extern.slf4j.Slf4j;


import java.util.concurrent.TimeUnit;
import java.util.concurrent.locks.*;


@Slf4j(topic = "enjoy")
public class LockTest10 implements Lock{
    CustomSync customSync = new CustomSync();


    @Override
    public void lock() {
        customSync.acquire(1);
```

```
    }

    @Override
    public void lockInterruptibly() throws InterruptedException {
        customSync.acquireInterruptibly(1);
    }

    @Override
    public boolean tryLock() {
        return customSync.tryAcquire(1);
    }

    @Override
    public boolean tryLock(long time, TimeUnit unit) throws InterruptedException
{
        return customSync.tryAcquireNanos(1, unit.toNanos(time));
    }

    @Override
    public void unlock() {
        customSync.release(1);
    }

    @Override
    public Condition newCondition() {
        return customSync.newCondition();
    }


    public static void main(String[] args) throws InterruptedException {
        LockTest10 l = new LockTest10();
        new Thread(()->{
            l.lock();
            log.debug("xxx");
            try {
                TimeUnit.SECONDS.sleep(5);
            } catch (InterruptedException e) {
                e.printStackTrace();
            }
            l.unlock();
        },"t1").start();

        TimeUnit.SECONDS.sleep(1);
        l.lock();
        log.debug("main");
        l.unlock();
    }
}
```

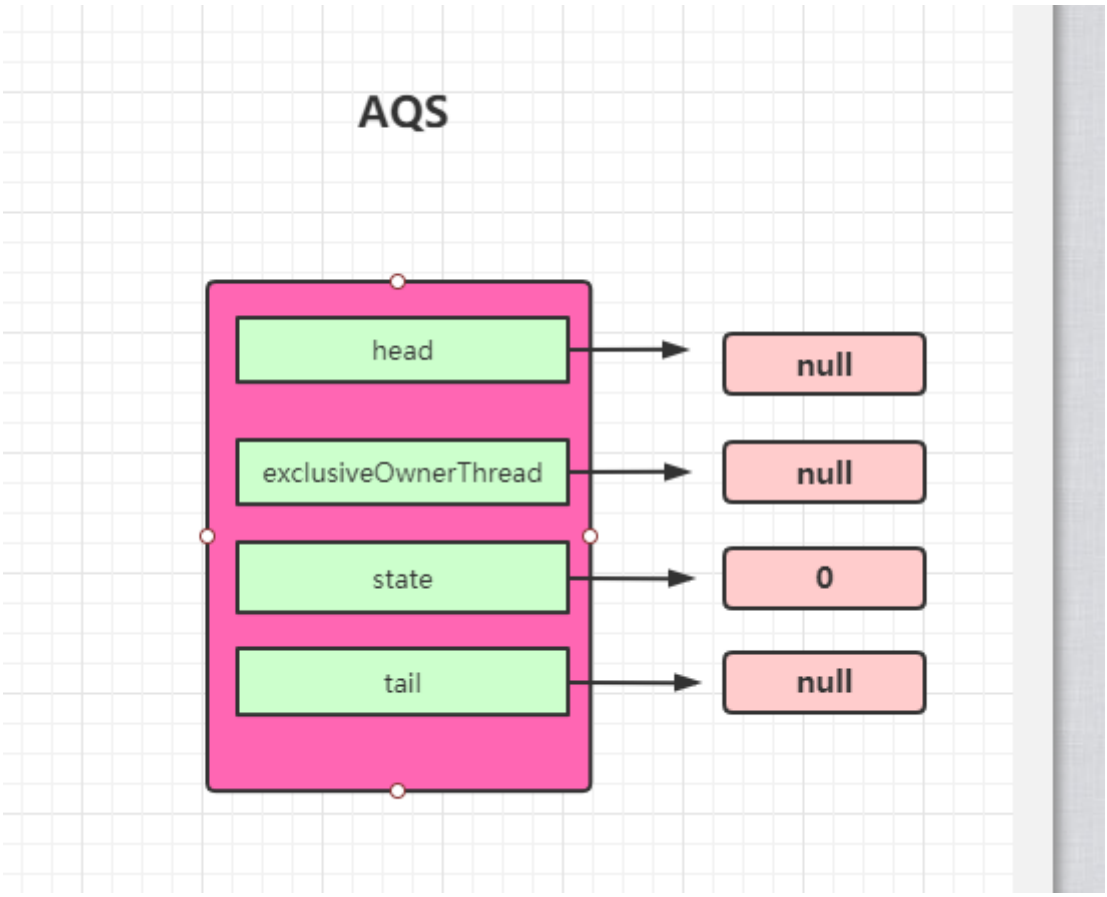
ReentrantLock

和Aqs的关系

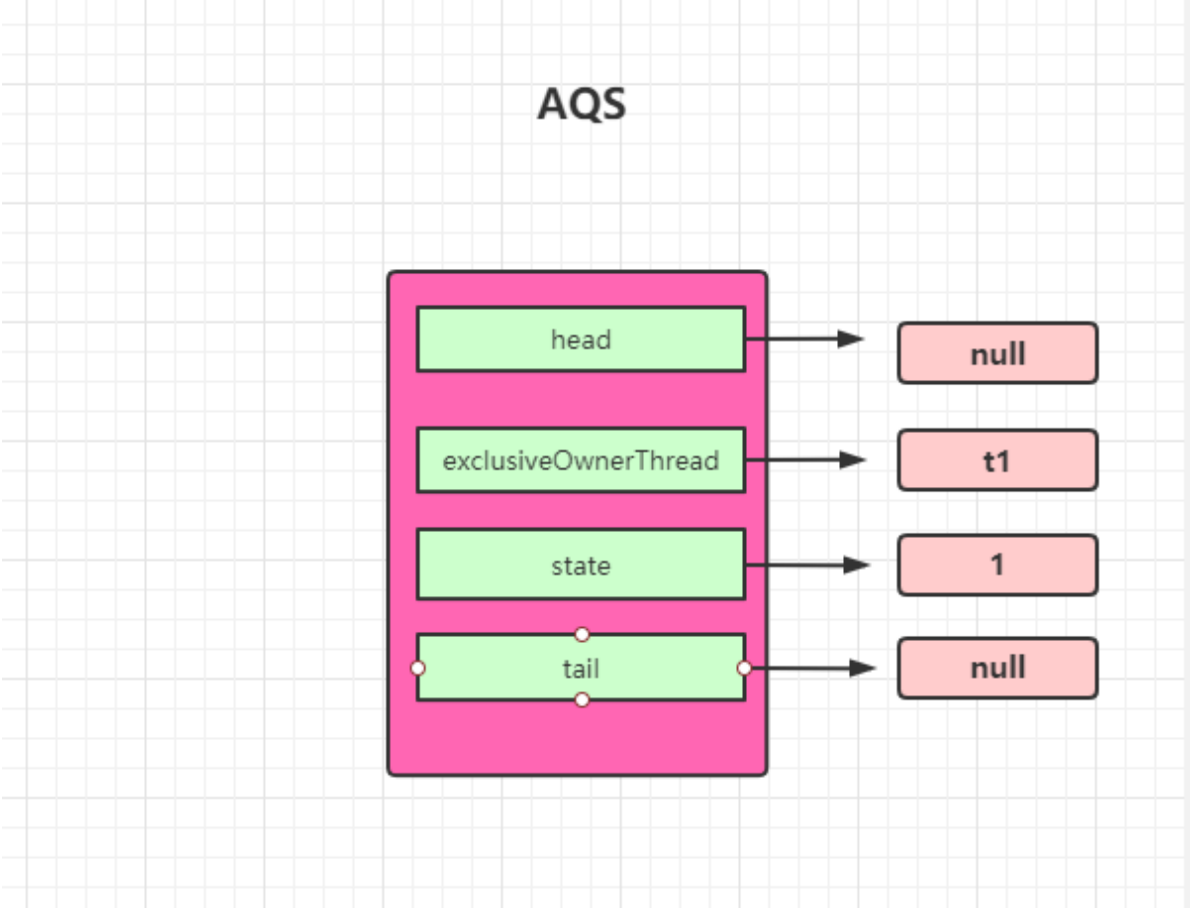


非公平锁加锁流程

1、第一个线程t1、第一次加锁，没有加锁之前 aqs（NonfairSync）的状态



2、t1、加锁之后



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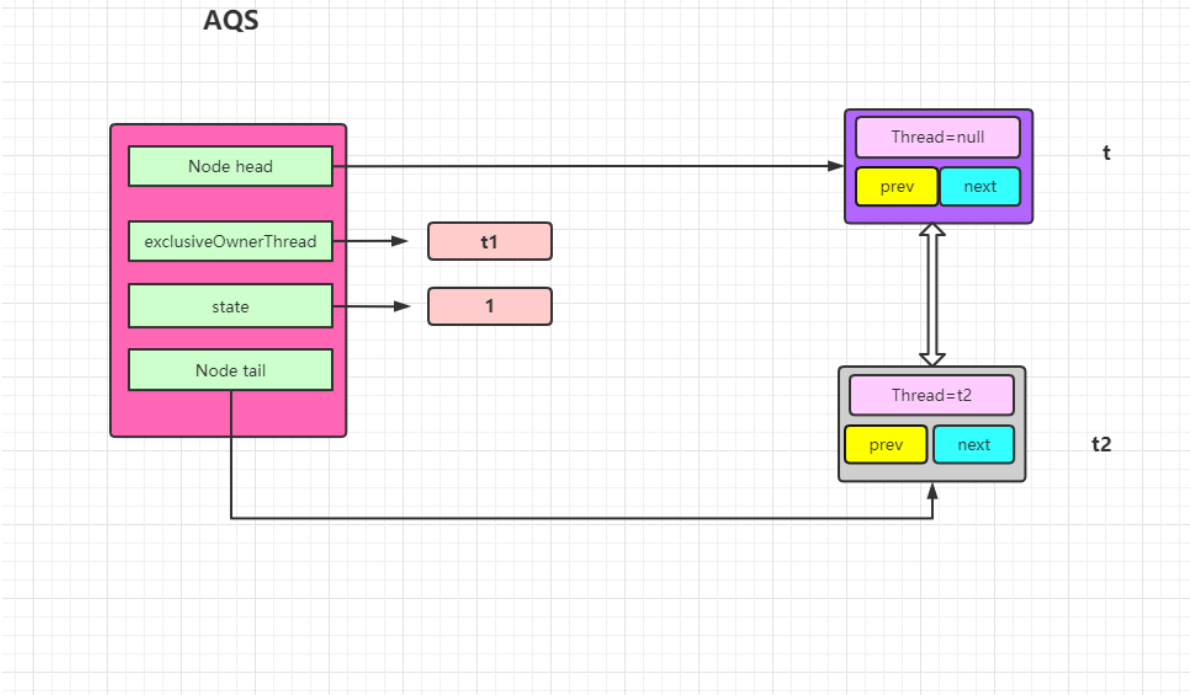
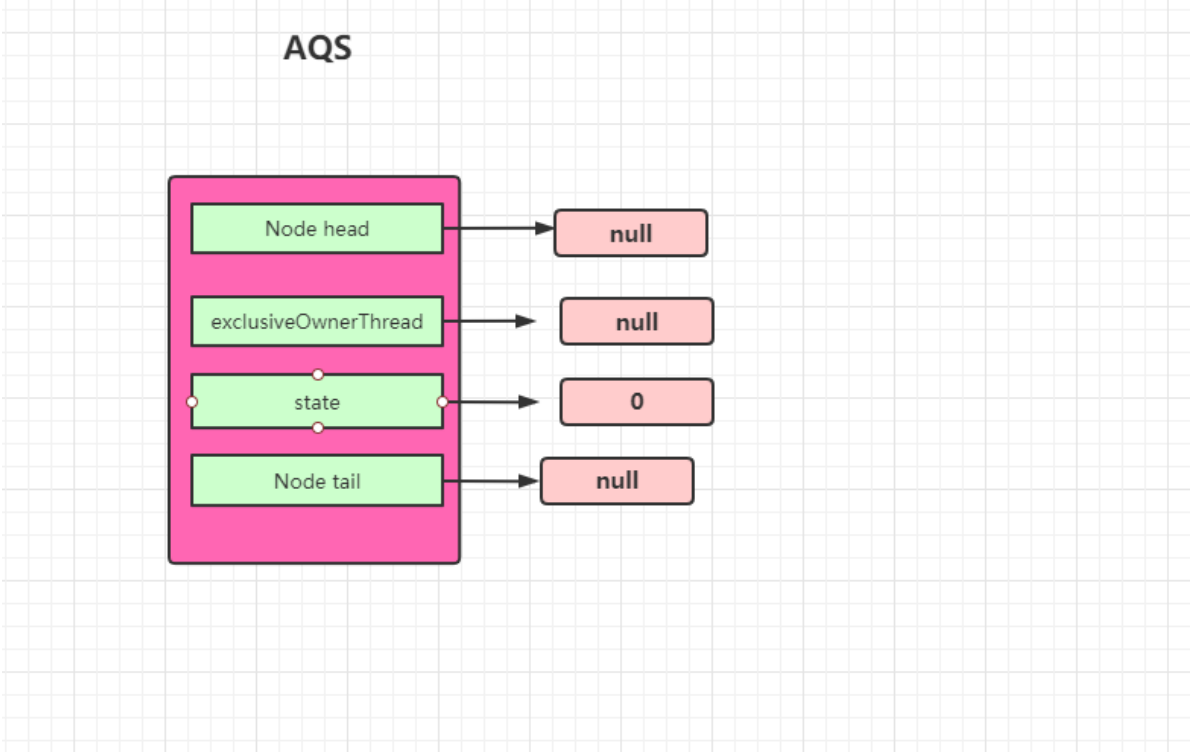
3、第二个线程t2 如果没有释放 AQS状态和2一样

3.1 t2 加锁之后失败之后

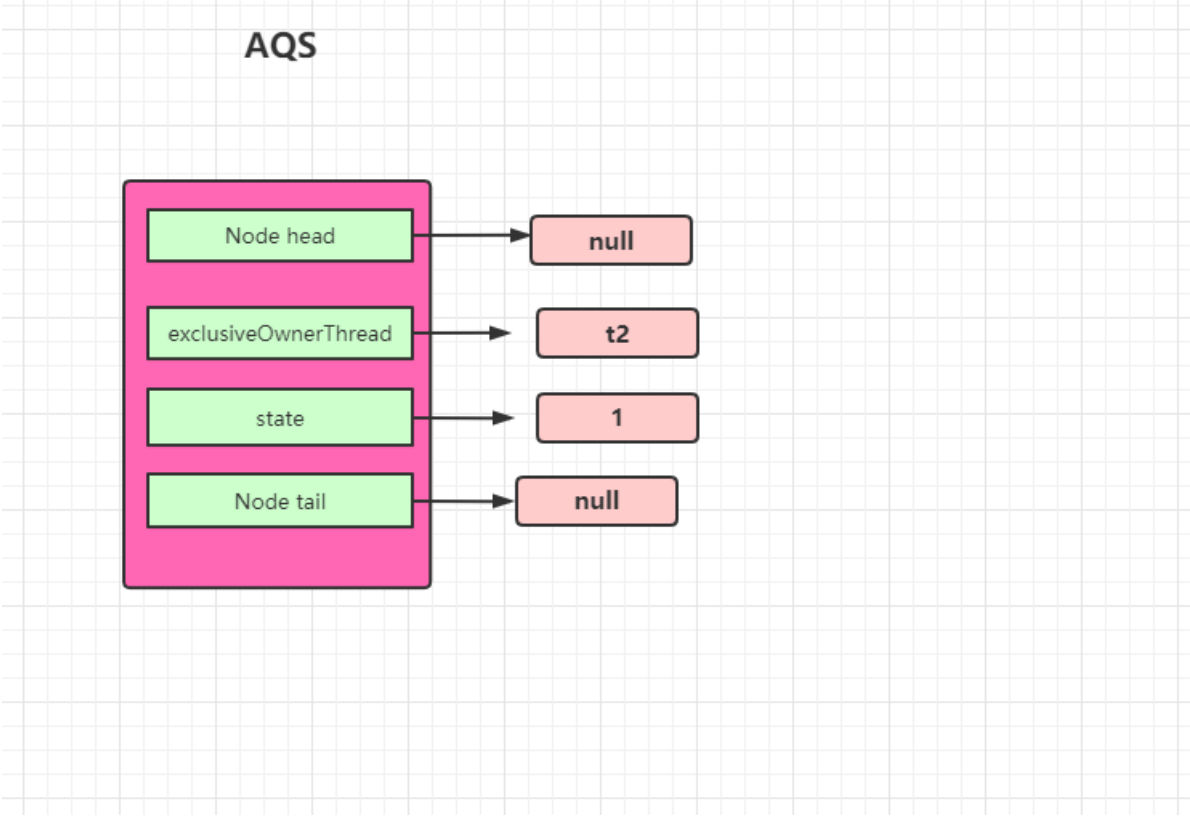


3.2 t2 加锁成功之后

如果需要加锁成功则t1必须释放锁

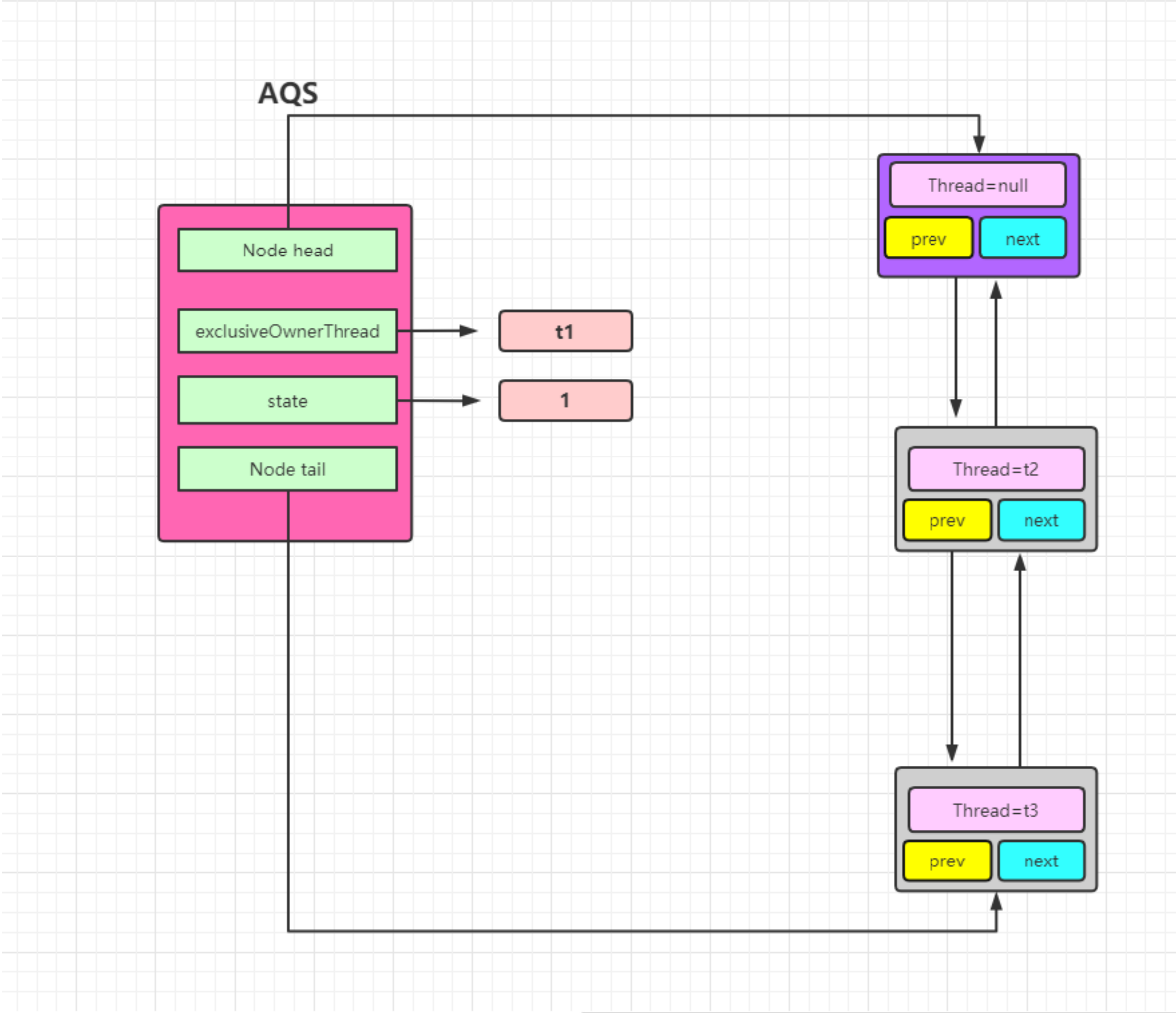


然后t2加锁成功



结论：ReentrantLock如果线程之间没有竞争效率非常高；甚至队列都没有初始化

4、t3来加锁


AQS
Node head
exclusiveOwnerThread
state
Node tail
t1
1
Thread=null
prev
next
Thread=t2
prev
next
Thread=t3
prev
next